

OLAP Data Modeling



Preparing data for mining...



University
of Calcutta

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Data Warehouse

- “A data warehouse is a *subject-oriented*, *integrated*, *time-variant*, *nonvolatile* collection of data in support of management’s decision making process”

- William H. Inmon



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Online Analytical Processing (OLAP)



- OLAP is a software technology that enables analysts, managers and executives to gain insight of data through fast, consistent, interactive access of *information that has been transformed from raw data* to reflect the real *dimensionality* of the enterprise as understood by the user

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OLAP VS OLTP



Feature	OLTP	OLAP
Characteristics	Operational processing	Informational processing
Orientation	Transaction	Analysis
User	Clerk, DBA, Operator	Knowledge worker, executive, analyst
Function	Day-to-day operations	Decision support
DB design	ER based, flat, application oriented	Multidimensional, subject oriented
Data	Current	Historical

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OLAP VS OLTP



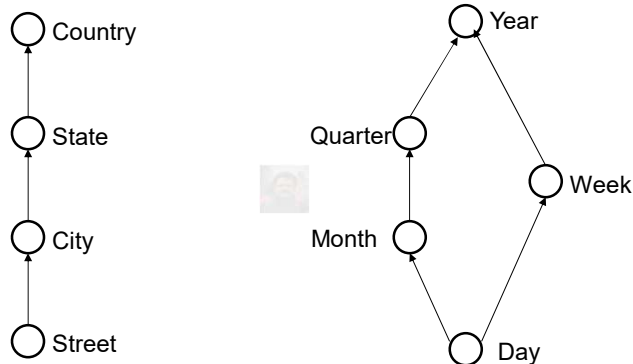
Feature	OLTP	OLAP
Summarization	Highly detailed	Consolidated
Unit of work	Short, simple transactions	Complex analysis
Access	Read/write	Mostly read
Records access per operation	10s	Millions
Number of users	Thousands	Hundreds
DB Size	100 MB to GB	100 GB to TB

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Concept Hierarchy



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Data Models for OLAP



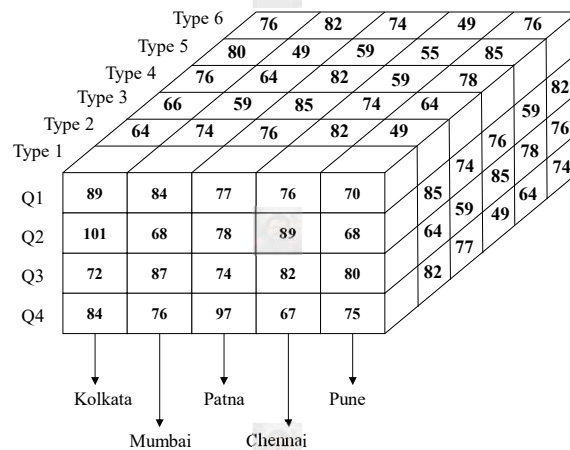
- Multi-dimensional Cube Data Model
 - Fact
 - Dimension
- Constellation Data Model
- Snowflake Data Model

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Cube Data Model



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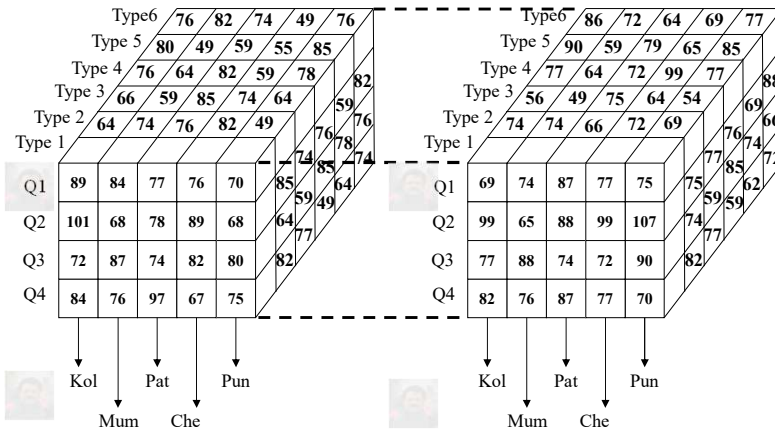
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4-D Cube Data Model



Fantaloan

Bug Bazar



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OLAP Operations



- Roll Up
- Drill Down
- Slice
- Dice
- Pivot



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Roll Up and Drill Down



- Drill-down refers to the process of viewing data at a level of increased detail
- Roll-up refers to the process of viewing data with decreasing detail
- Drilling down or up is a specific analytical technique whereby the user navigates among levels of data ranging from the most summarized (up) to the most detailed (down)

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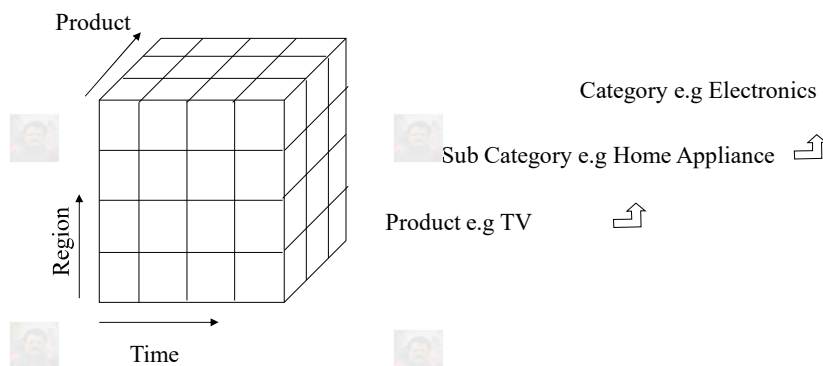
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OLAP Operations



Roll Up



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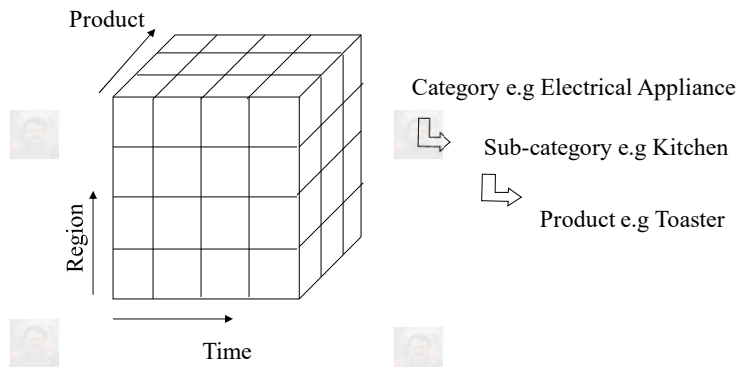
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OLAP Operations



Drill Down



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Slice



- A slice is a subset of a multi-dimensional array corresponding to a single value for one or more members of the dimensions not in the subset.

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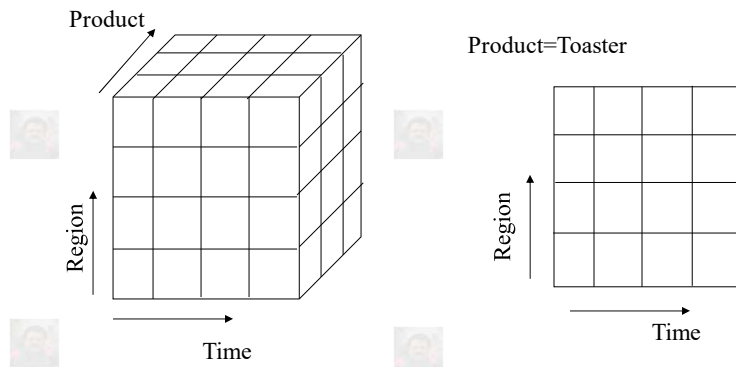
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OLAP Operations



Slice



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Dice



- Dice operation is done to select two or more dimensions from a given cube to yield a new sub-cube.
- The dice operation is analogous to slice on multiple dimensions of a data cube, or more than values on the same dimension.

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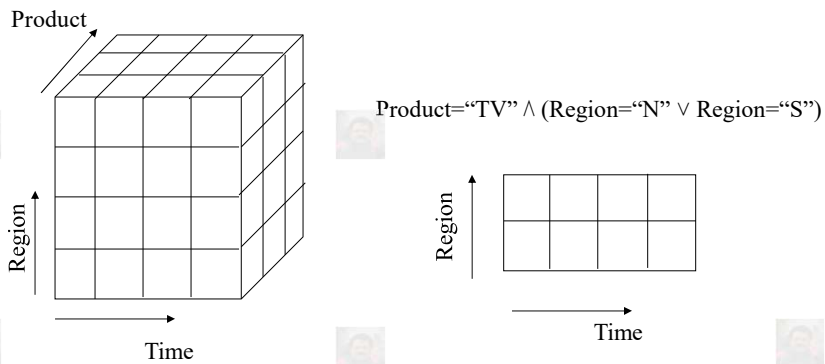
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OLAP Operations



Dice



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Pivot



- Pivot is another operation in OLAP to change the dimensional orientation for an alternate view of the data cube

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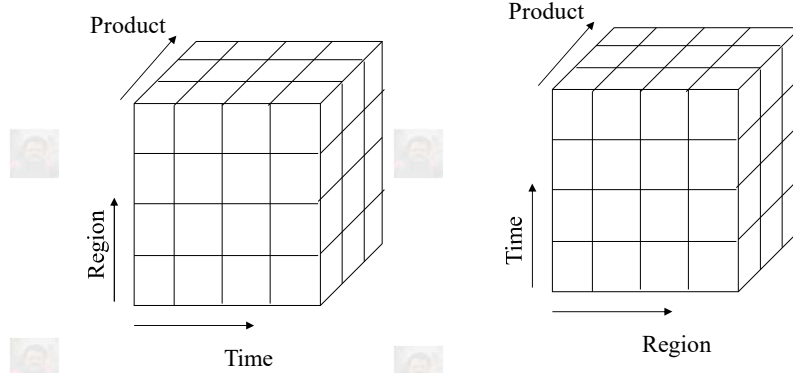
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OLAP Operations



Pivot



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Data Warehouse Schema



- Star Schema
- Fact Constellation Schema
- Snowflake Schema

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Star Schema



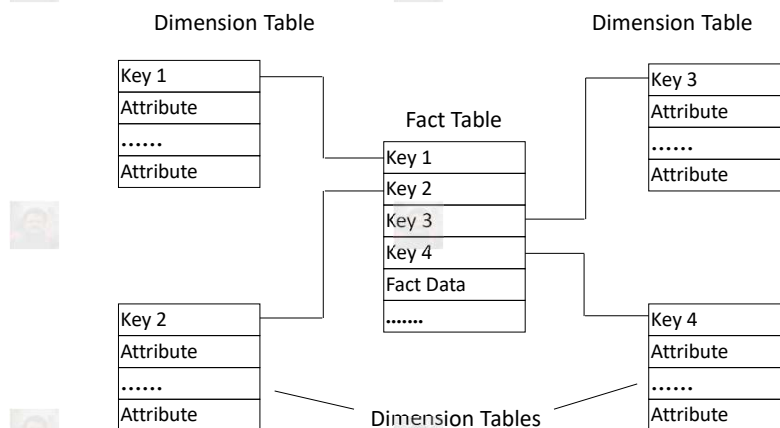
- A single, large and central fact table and one table for each dimension.
- Every fact points to one tuple in each of the dimensions and has additional attributes.
- Does not capture hierarchies directly.

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Star Schema



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Star Schema: Example



Customer

CustID
Name
.....
Phone

Shop

ShopID
Location
.....
Owner

Sales

CustID
ProdID
ShopID
CompID
Quantity
Turnover

Product

ProdID
ItemName
.....
Category

Company

CompID
CompName
.....
Status

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Star Schema: Example



CustID	Name	Phone
C1	Amit	1234
C2	Biman	2345

ProdID	ItemName	Category
P1	Shirt	2
P2	Camera	4
P3	Watch	4

CustID	ProdID	ShopID	CompID	Quantity	Turnover
C1	P1	S1	CM1	100	2500
C2	P1	S2	CM1	140	2300
C1	P1	S3	CM2	180	1950
C1	P3	S1	CM3	160	2600

ShopID	Location	Owner
S1	Kasba	PQR
S2	Patuli	JKL

CompID	CompName	Status
CM1	XYZ	1
CM2	ABC	1

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Snowflake Schema



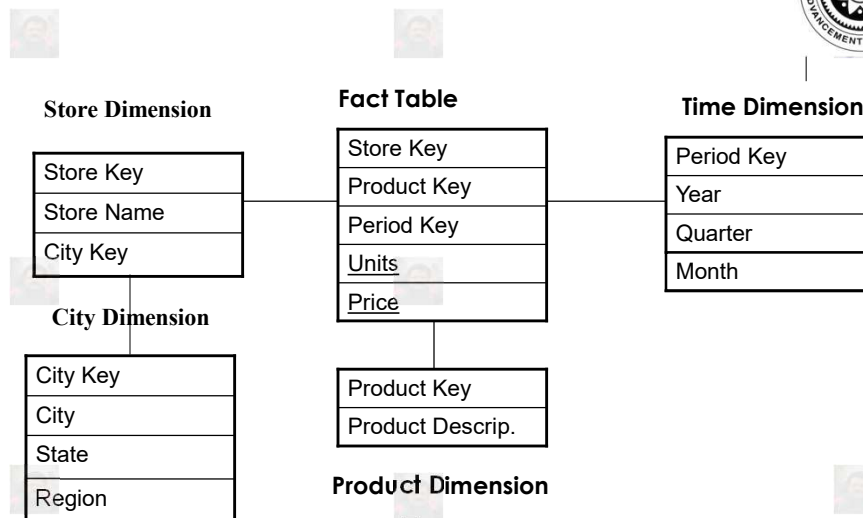
- Variant of Star schema model.
- A single, and central fact table and one or more tables for each dimension.
- Dimension tables are normalized i.e., split dimension table data into additional tables

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Snowflake Schema



Drawbacks: Time consuming joins, report generation slow

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Fact Constellation



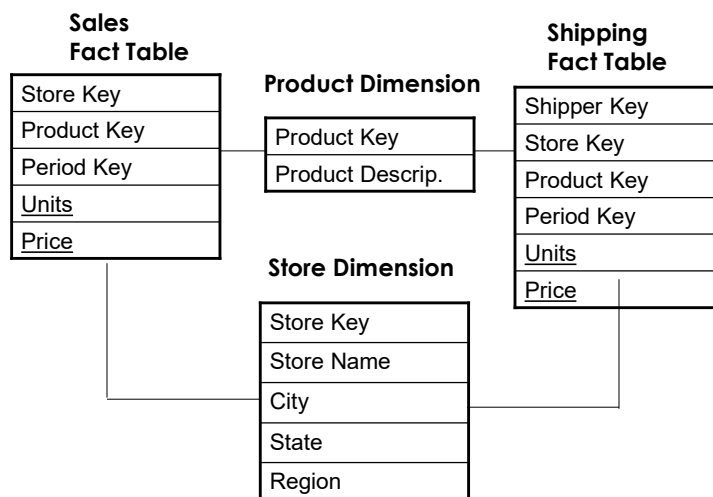
- Multiple fact tables share dimension tables.
- This schema is viewed as collection of stars hence called galaxy schema or fact constellation.
- Sophisticated application requires such schema.

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Fact Constellation



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Aggregation



- Add up amounts for day 1
- In SQL: `SELECT sum(amt) FROM SALE WHERE date = 1`

sale	prodId	storeId	date	amt
	p1	c1	1	12
	p2	c1	1	11
	p1	c3	1	50
	p2	c2	1	8
	p1	c1	2	44
	p1	c2	2	4

➔ 81

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Aggregation



- Add up amounts by day
- In SQL: `SELECT date, sum(amt) FROM SALE GROUP BY date`

sale	prodId	storeId	date	amt
	p1	c1	1	12
	p2	c1	1	11
	p1	c3	1	50
	p2	c2	1	8
	p1	c1	2	44
	p1	c2	2	4



ans	date	sum
	1	81
	2	48

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Aggregation



- Add up amounts by day, product
- In SQL: `SELECT date, sum(amt) FROM SALE GROUP BY date, prodId`

sale	prodId	storeId	date	amt
	p1	c1	1	12
	p2	c1	1	11
	p1	c3	1	50
	p2	c2	1	8
	p1	c1	2	44
	p1	c2	2	4



sale	prodId	date	amt
	p1	1	62
	p2	1	19
	p1	2	48

rollup →

← drill-down

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Major References



- Jiawei Han, Micheline Kamber; "Data Mining: Concepts and Techniques" 2nd Ed., Morgan Kaufmann Series in Data Management Systems; Elsevier Science; 2006; ISBN: 978-1-55860-901-3.
- William H Inmon; "Building Data Warehouse"; QED Technical Publications Group; 1992; ISBN: 978-0-89435-404-5.

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